



L3 In-Lecture Questions

Q1: Gradient direction

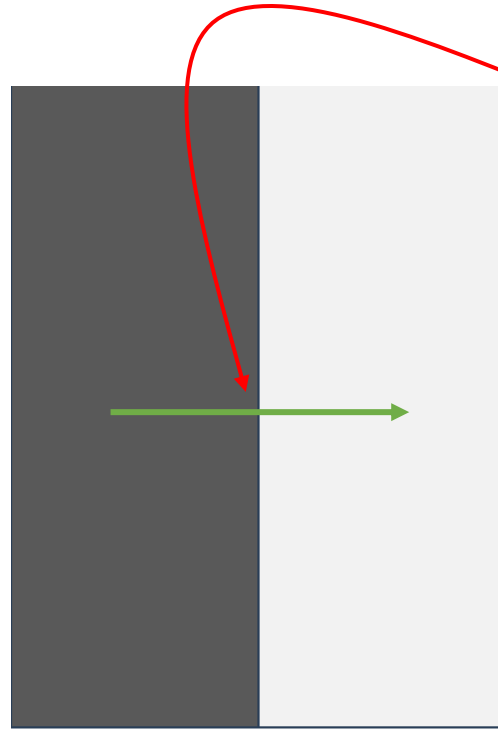
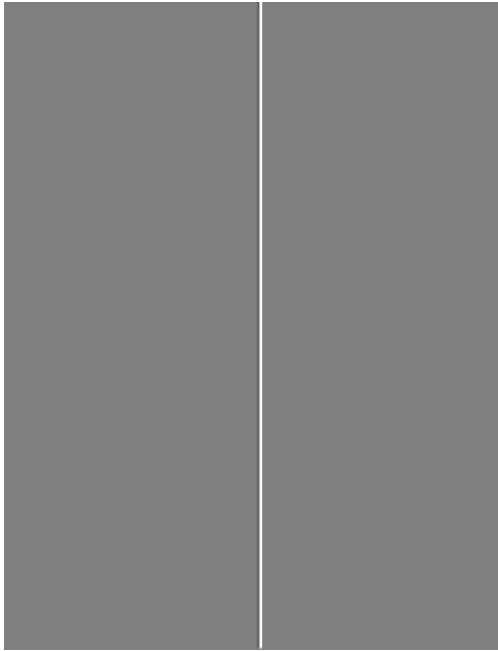


A dark-to-bright vertical edge is present in an image. Which statement is correct?

- A. G_x will be strongest, and the gradient points across the edge.
- B. G_x will be strongest, and the gradient points along the edge.
- C. G_y will be strongest, and the gradient points along the edge.
- D. G_x and G_y must have equal magnitudes.

A1: Gradient direction (A)

Gx captures vertical edges



Gradients measure the change in intensity along a direction

Gx will be strongest, and the gradient points across the edge.

Q2: First versus second derivatives

For an **ideal intensity step** that has first been smoothed, where is the edge located?

- A. At a zero-crossing of the first derivative
- B. At a maximum of the first-derivative magnitude
- C. At a maximum of the Laplacian magnitude
- D. Where both the first derivative and Laplacian are zero

A2: First versus second derivatives (B)

What is an **ideal intensity step**?



An immediate change between 2 constant regions

What does it look like smoothed?



The edge will be blurred

- A. First derivative is 0 in flat regions
- B. The edge is where intensity changes most rapidly, so the absolute value of the first derivative is largest there.**

- C. It would have a positive lobe on one side and a negative lobe on the other. The actual edge center lies between those lobes.
- D. At the edge center, the Laplacian can be zero, but the first derivative is actually at its maximum magnitude, not zero..

Q3: Sobel filter structure

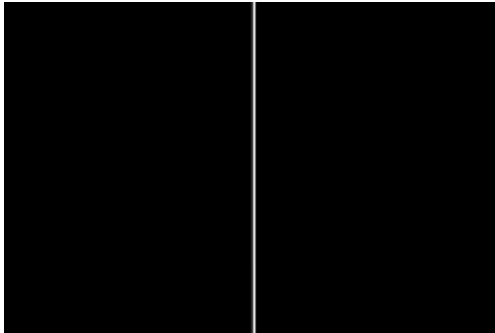
$$G_x = \begin{array}{|c|c|c|} \hline -1 & 0 & 1 \\ \hline -2 & 0 & 2 \\ \hline -1 & 0 & 1 \\ \hline \end{array}$$

$$G_y = \begin{array}{|c|c|c|} \hline 1 & 2 & 1 \\ \hline 0 & 0 & 0 \\ \hline -1 & -2 & -1 \\ \hline \end{array}$$

Why does the Sobel x-filter contain negative weights on one side and positive weights on the other?

- A. To compute the difference between intensities on opposite sides
- B. To ensure that all output values remain between 0 and 255
- C. To detect only bright lines on dark backgrounds
- D. To make the filter invariant to edge orientation

A3: Sobel filter structure (A)



Correct Gx filter



Abs(Gx) filter

- A. The negative and positive weights effectively compare the brightness on the left and right sides. A large difference means there is likely a vertical edge.
- B. Sobel can be negative or positive, indicating the direction of change
- C. Sobel can detect both dark-to-bright and bright-to-dark edges.
- D. The Sobel x-filter is specifically sensitive to changes in the x-direction, so it is not invariant to edge orientation.

Q4: Canny Edge Detection



Which Canny step is primarily responsible for producing thin edges?

- A. Gaussian smoothing
- B. Point interpolation
- C. Non-maximum suppression
- D. Hysteresis thresholding

A4: Canny Edge Detection (C)

- A. It can make the gradient more stable, but its main purpose is noise reduction, not thinning edges.
- B. Interpolation may be used within non-maximum suppression to estimate gradient values between pixels, but it is not itself the main Canny step responsible for thinning.
- C. Neighboring weaker responses are suppressed, turning broad edge responses into thin, approximately one-pixel-wide edges.**
- D. This decides which detected edges to keep. Strong edges are retained, and weak edges are retained only when connected to strong edges.